

Emotion Dynamics Project

Research Report

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Abstract

Emotion Recognition in Conversation (ERC) aims to identify the emotional state of an utterance by considering both its linguistic content and conversational context. This project investigates how different utterance representations and contextual modeling strategies affect BERT-based ERC on the DailyDialog dataset. Starting from a single-utterance BERT baseline, the study progressively examines explicit conversational context, recurrent aggregation, Transformer-based self-attention, and structured attention masks designed to capture global, local, inter-speaker, and intra-speaker relationships.

During development, a batching implementation error was identified that had affected a substantial portion of the initial contextual experiments. After correcting the error, the affected experiments were repeated under a consistent training procedure. The final contextual architectures achieved performance comparable to the single-utterance baseline, but did not provide consistent evidence of a meaningful improvement. Analysis of the resulting contextual representations further showed substantial similarity between several attention-mask branches, suggesting that additional architectural structure does not necessarily yield substantially different or more task-relevant information. Overall, the study provides an empirical investigation of contextual modeling in BERT-based ERC while highlighting the importance of strong baselines, implementation verification, controlled experimentation, and representation-level analysis. The implementation and experimental configurations are available at <https://github.com/Vu-Hoang-Anh-Quan/Emotion-Dynamics-Project>.

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1 Introduction

Emotion Recognition in Conversation (ERC) is the task of identifying the emotional state expressed in an utterance while considering the surrounding conversational context. Unlike conventional emotion classification, where an utterance can often be treated independently, ERC requires a model to account for information distributed across multiple turns of a dialogue. The interpretation of an utterance may depend on preceding statements, speaker identity, and the broader conversational structure.

Consequently, ERC systems have explored various approaches to incorporating conversational context, including recurrent architectures, attention mechanisms, graph-based models, and pre-trained language representations. However, increasing the amount or complexity of contextual information does not necessarily improve emotion recognition. A strong single-utterance representation may already capture substantial information, while additional context may be redundant or introduce irrelevant information. This makes the manner in which contextual relationships are represented and integrated an important experimental question.

This project investigates the empirical effects of different representation, context-integration, and architectural choices in BERT-based ERC on the DailyDialog dataset. The study progresses from single-utterance baselines to increasingly structured contextual models, examining BERT pooling strategies, explicit conversational context, recurrent aggregation, self-attention, and structured attention masks based on temporal and speaker relationships. In addition to classification performance, the study analyzes the similarity between representations produced by different attention mechanisms to examine whether structurally different contextual views actually encode distinct information.

The experiments ultimately show that the more complex contextual architectures achieve performance comparable to, but not consistently better than, the single-utterance baseline. The research process also exposed an implementation error in the batching procedure that affected a substantial portion of the initial contextual experiments. After correcting the error, the affected experiments were repeated and the analysis was revised accordingly. Together, these experiments provide an empirical investigation of the relationship between utterance representation, contextual modeling, architectural structure, and experimental reliability in BERT-based ERC.

2 Background

Emotion Recognition in Conversation (ERC) is an established research problem in natural language processing that focuses on predicting the emotional state of individual utterances within a conversational context. The development of ERC has been supported by conversational datasets containing utterance-level emotion annotations, such as DailyDialog, which provides multi-turn dialogues together with manually annotated emotion and communication-intention labels [Li et al., 2017]. This formulation distinguishes ERC from conventional text classification because the input relevant to an emotion prediction is not necessarily limited to the target utterance itself.

Early ERC approaches consequently incorporated conversational history into the modeling process. For example, the Conversational Memory Network maintained separate memories for the speakers and used attention to model their interactions, demonstrating the importance of con-

textual and inter-speaker information for emotion recognition [Hazarika et al., 2018]. Subsequent recurrent approaches further modeled the evolution of utterance representations throughout a dialogue. For example, DialogueRNN explicitly models the emotional states of individual speakers and their evolution across conversational turns [Majumder et al., 2019]. These approaches established a recurring theme in ERC research: the representation of an utterance can benefit from information contained in other utterances, but the effectiveness of such information depends on how conversational dependencies are modeled.

Research subsequently moved beyond treating context as a simple sequential history and began modeling more explicit relationships between utterances. DialogueGCN, for example, represents a conversation as a graph containing self-speaker and inter-speaker dependencies, allowing information to propagate through explicitly defined conversational relationships [Ghosal et al., 2019]. Other approaches employed hierarchical or attention-based mechanisms to determine which contextual utterances should contribute to the representation of a target utterance. Knowledge-Enriched Transformer, for instance, combines hierarchical self-attention over contextual utterances with commonsense knowledge, while DialogueCRN explicitly introduces contextual reasoning for emotion recognition [Zhong et al., 2019, Hu et al., 2021]. These developments reflect a broader shift from simply adding conversational history to explicitly modeling interactions within that history.

Another direction has been to enrich contextual representations with information about the causes, mental states, and interactions underlying conversational emotions. COSMIC incorporates commonsense representations of events, mental states, and causal relations to model interactions between interlocutors [Ghosal et al., 2020]. More recent work continues to extend this idea by introducing additional speaker-level or psychological information into contextual modeling [Lee and Lee, 2022, Wang et al., 2024]. Thus, contemporary ERC research encompasses several complementary sources of information, including linguistic content, temporal context, speaker relationships, commonsense knowledge, and inferred psychological states.

Within this development, pretrained language models have also become a natural foundation for representing individual utterances. BERT provides contextual token representations learned from large-scale pre-training and can be fine-tuned for downstream classification tasks [Devlin et al., 2019]. Consequently, an ERC architecture can be viewed as consisting of two conceptually distinct components: an *utterance representation* that captures the linguistic content of an individual turn, and a *contextual modeling mechanism* that determines how representations of different turns interact.

The distinction between these two components is particularly relevant to the present work. Improvements in the representation of an individual utterance do not by themselves determine how information from the surrounding conversation should be incorporated. Conversely, a contextual model may have access to substantial conversational information while still being limited by how it defines or aggregates relationships between utterances. Existing ERC research demonstrates the value of contextual information through recurrent, attention-based, graph-based, and knowledge-aware architectures, but these approaches make different assumptions about which utterances should interact and how those interactions should occur [Ghosal et al., 2019, Zhong et al., 2019, Ghosal et al., 2020].

This motivates treating *contextual relationship modeling* as a central architectural consideration in ERC. Rather than assuming that all contextual information is equally useful, the relevant question is how the relationships among utterances should be represented and controlled when constructing a contextualized representation. Temporal proximity, speaker identity, and the

direction of conversational information flow provide different possible structures for these interactions. Understanding the effect of such structures provides a focused way to study contextual modeling independently from the increasingly broad set of external knowledge and multimodal signals explored in ERC.

3 Objectives

The primary objective of this project is to investigate how contextual relationships between utterances can be modeled to improve emotion recognition in conversations. Specifically, the project aims to:

1. Establish a reliable single-utterance baseline using BERT and examine how individual utterance representations affect emotion recognition performance.
2. Investigate whether alternative representations of an utterance, such as mean pooling over token representations, provide more suitable representations for conversational emotion recognition.
3. Determine whether explicitly incorporating information from surrounding utterances improves emotion recognition compared with modeling each utterance independently.
4. Compare different approaches to contextual aggregation, particularly recurrent and attention-based architectures, and examine how they affect the representation of conversational context.
5. Investigate how restricting attention through different attention masks influences the contextual relationships captured between utterances, with particular focus on temporal, speaker, and listener relationships.
6. Examine how architectural and training choices affect the quality of the resulting representations and the overall emotion recognition performance.
7. Analyze the experimental results to identify which types of contextual relationships are most useful for modeling emotional states in conversations.

4 Research Process

4.1 Problem Formulation and Initial Hypothesis

Emotion Recognition in Conversation (ERC) requires predicting the emotional state of an utterance while considering its conversational context. I initially hypothesized that contextual information would improve emotion recognition compared with treating each utterance independently. Therefore, the first stage of the project focused on establishing a single-utterance baseline before progressively introducing more complex forms of context modeling.

4.2 Establishing the Single-Utterance Baseline

I first fine-tuned BERT to classify each utterance independently. The initial implementation used the representation of the [CLS] token as the utterance embedding. These embeddings then go through the classifier,

$$pred = \arg \max(\text{Softmax}(H_{[CLS]}W))$$

This model achieved a macro F1-score excluding the Neutral label of 0.3309 on the validation set and 0.3643 on the test set. This metric will be referred to as the F1-score throughout the remainder of this report.

4.3 Introducing context

Having established the single-utterance baseline, I next investigated whether conversational context could provide additional information for emotion recognition. For each target utterance u_t , the initial contextual approach formed the input to BERT by concatenating the k preceding utterances with the current utterance:

$$x_t = u_{t-k} \parallel u_{t-k+1} \parallel \dots \parallel u_{t-1} \parallel u_t, \quad h_t = \text{BERT}(x_t),$$

where $\parallel$ denotes sequence concatenation. Setting $k = 3$ increased the validation F1-score to 0.3910 and the test F1-score to 0.4314.

Although this result suggested that conversational context could be beneficial, concatenating utterances directly into a single sequence limited the model’s ability to explicitly distinguish and selectively incorporate information from different parts of the conversation. I therefore considered an utterance-level architecture in which each utterance retained its own representation and could selectively attend to other utterances in the conversation.

4.4 Unexpected performance and Debugging

4.4.1 Gated Recurrent Unit Networks

To introduce conversational context while preserving the independently learned utterance representations, I first used a Gated Recurrent Unit Networks (GRU) as a context aggregation module. Given the sequence of utterance representations produced by BERT, $H = [h_1, h_2, \dots, h_T]$, the GRU recursively processed the sequence and produced contextualized representations, $z_t = \text{GRU}(h_t, z_{t-1})$.

The resulting representation was then passed to the emotion classifier. I chose the GRU primarily because it provided a relatively simple mechanism for aggregating contextual information while preserving the ordering of utterances.

However, the resulting performance was substantially worse than the single-utterance baseline, achieving F1-score of 0.0863 on validation set and 0.0908 on test set. I initially hypothesized that the GRU might be disrupting useful information already encoded by BERT. In particular, since gradients from the emotion classification objective propagated through the GRU before reaching

BERT, I suspected that the additional recurrent transformation could produce undesirable gradients and consequently degrade the representations learned by BERT.

4.4.2 Self-attention

To investigate whether the problem was specific to recurrent context aggregation, I replaced the GRU with a self-attention mechanism. Instead of recursively processing the utterances, self-attention computes contextualized representations through interactions among all available utterances:

$$\begin{aligned} Q &= HW_Q, & K &= HW_K, & V &= HW_V, \\ A &= \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + M \right), \\ H' &= AV. \end{aligned}$$

4.4.3 Mean pooling

At this point, I considered whether the representation produced by BERT itself was contributing to the instability. The initial model used the [CLS] representation, $h_{\text{CLS}} = H_{[\text{CLS}]}$. This representation, in BERT, is originally used for next-sentence predicting, not specifically for emotional detecting. I suspected that operating on CLS to act as utterance representations might capture strong semantics but limited emotion from each token.

Therefore, I replaced [CLS] pooling with mean pooling,

$$\frac{1}{N} \sum_{i=1}^N h_i,$$

where h_i denotes the representation of the i -th non-padding token. The resulting utterance representation was then provided to the self-attention and classifier module.

The architecture achieved 0.1045 on the validation set and 0.1005 on the test set.

4.4.4 Separated training

I next attempted to isolate the effect of joint optimization between BERT and the contextual module. First, BERT was fine-tuned independently for single-utterance emotion recognition. After training, its classification head was removed, leaving the resulting utterance encoder,

$$h_t = \text{BERT}_\theta(u_t).$$

The encoder was then connected to the self-attention module and a new classifier,

$$H = [h_1, \dots, h_T],$$

$$H' = \text{SelfAttention}(H),$$

$$\hat{y}_t = f_\phi(H'_t).$$

During this second stage, the BERT parameters were frozen so that only the contextual module and classifier were optimized. This experiment was intended to determine whether the contextual module could improve an already effective single-utterance representation without simultaneously modifying BERT.

The independently trained BERT model achieved substantially stronger performance than the original baseline, achieving approximately 0.36 on the validation set and 0.39 on the test set. However, once the frozen BERT representations were passed through the contextual module, the resulting performance again dropped substantially to around 0.12 on both validation and test set. This indicated that the poor performance was unlikely to be explained solely by undesirable gradients flowing from the contextual module into BERT.

Throughout these experiments, I also encountered numerical instability during training. In particular, NaN values occasionally appeared within the attention computation, with the instability sometimes being localized to individual elements of larger parameter or activation tensors. I investigated several possible sources, including mixed-precision training and the numerical behavior of the attention computation. The precise cause of these localized instabilities could not be conclusively identified. However, switching the training procedure to bfloat16 (bf16) eliminated the observed NaN behavior, allowing subsequent experiments to proceed.

4.5 Corrected Attention Experiments

Before identifying the batching error, I conducted several additional experiments to determine whether the poor performance originated from specific components of the attention architecture. These experiments investigated the residual connection, the attention behavior itself, and the capacity of the contextual module.

4.5.1 Residual Connections

I first investigated the residual connection. The initial implementation used the standard residual formulation,

$$H_{\text{out}} = H_{\text{attn}} + X,$$

where X denotes the input utterance representations and H_{attn} denotes the output of the attention module.

I also tested concatenating rather than adding the residual information,

$$H_{\text{out}} = \text{Concat}(H_{\text{attn}}, X)W_r.$$

None of these modifications resolved the poor performance. This suggested that the residual pathway itself was unlikely to be the primary cause of the observed degradation.

4.5.2 Attention Distribution and Self-Bias

I next investigated the attention distributions directly. Inspection of the attention scores revealed an unexpected pattern: for most of the observed attention distributions, the probability assigned to the current utterance was at most 0.5, and was typically around 0.3. The remaining probability was distributed relatively evenly among other utterances.

However, I observed that in ERC, the information required to determine an utterance’s emotion often lies in approximately 2 utterances, usually including the current considering utterance and at most 2 other utterances (without regard of distance).

This led me to suspect that the attention mechanism might be failing to preserve sufficient information from the current utterance while aggregating contextual information.

I therefore introduced a learnable self-bias to explicitly encourage attention toward the current utterance. In addition, I tested a fixed bias designed to enforce a stronger preference for the current utterance. In simplified form, the attention logits were modified as

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} + M + B \right),$$

where B represented either a learnable or fixed self-bias. Both approaches resulted in substantially worse performance, with F1-scores of approximately 0.10 on the validation set and 0.12 on the test set. Therefore, the observed attention distribution did not appear to be simply a consequence of insufficient self-attention to the current utterance.

4.5.3 Attention Distribution Regularization

Since the attention distributions appeared to spread their probability across many utterances, I next introduced an auxiliary regularization loss to encourage the model to attend primarily to a small local context. This was motivated by the observation that, in ERC, the information required to determine an utterance’s emotion often lies within approximately two surrounding utterances, although the relevant context may vary between one and three utterances.

I quantified the concentration of each attention distribution using the Rényi entropy of order two,

$$H_2 = -\log \left(\sum_i p_i^2 \right),$$

and encouraged it to approach $\log 2$, corresponding approximately to a distribution concentrated over two utterances:

$$\mathcal{L}_{\text{reg}} = (H_2 - \log 2)^2.$$

The final objective was

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{reg}}.$$

However, this modification did not improve the model’s performance. The validation and test F1-scores remained approximately 0.10 and 0.12, respectively, suggesting that explicitly constraining the attention distribution to the expected contextual range was insufficient to resolve the underlying problem.

4.5.4 Increasing Attention Capacity

I subsequently investigated whether the attention module lacked sufficient representational capacity. The original architecture used a single attention operation, so I replaced it with multi-head self-attention:

$$\text{Attention}(HW_i^Q, HW_i^K, HW_i^V),$$

$$\text{Concat}(\text{head}_1, \dots, \text{head}_m)W^O.$$

The motivation was that different heads could potentially capture different relationships between utterances and provide richer contextual information than a single attention representation.

Despite these architectural modifications, the model continued to exhibit unexpectedly poor performance. At this point, the repeated failure of substantially different architectural variants made it increasingly unlikely that the problem could be attributed to a single design choice within the attention module. I therefore returned to the implementation and training pipeline and continued debugging the experiments.

4.5.5 Identification of the Batching Error

During this investigation, I suddenly found that changing the batch size at each experiment changed the score substantially, with larger batch size resulted in worse result. I eventually identified an error in the batching procedure that affected the contextual experiments. This

explained the anomalous behavior observed across the preceding experiments and meant that their absolute performance could not be reliably interpreted as evidence against the corresponding architectural choices. I therefore discarded the affected results and repeated the attention experiments using the corrected batching procedure.

4.6 Corrected Attention Experiments

With the corrected implementation, mean pooling, separated training pipeline, and the multiple head attention module, the model achieved performance comparable and slightly stronger than the mean pooling single-utterance detection baseline, achieving 0.3803 on the validation set and 0.4375 on the test set, while the mean-pooling baseline achieving 0.3488 and 0.4363 on the validation and test set, respectively. This shifted the investigation from the question of whether the attention architecture was functioning correctly to whether it was actually extracting information that was useful for the ERC task.

I hypothesized that self-attention, despite being capable of modeling interactions across the conversation, might not sufficiently distinguish which contextual relationships were relevant to the target utterance so that it can perform noticeably better than the baseline. This motivated the subsequent investigation of explicitly structured attention masks.

4.7 Exploring Multiple Contextual Masks

To investigate whether explicitly separating different types of contextual relationships could provide more useful information, I introduced four structured attention masks corresponding to global, local, inter-speaker, and intra-speaker views:

$$\mathcal{M} = \{M_{\text{global}}, M_{\text{local}}, M_{\text{inter}}, M_{\text{intra}}\}.$$

All four masks share a common base mask that combines causal and padding constraints. Using the convention that a value of 1 blocks an attention connection, the causal mask is defined as

$$M_{\text{causal}}(i, j) = \begin{cases} 1, & j > i, \\ 0, & j \leq i. \end{cases}$$

The padding mask prevents attention to invalid dialogue positions. The shared base mask is therefore

$$M_{\text{base}} = M_{\text{causal}} \vee M_{\text{padding}}.$$

The global mask applies only these common constraints, allowing each utterance to attend to every valid preceding utterance:

$$M_{\text{global}} = M_{\text{base}}.$$

The local mask further restricts attention to a window containing the target utterance and its $k - 1$ most recent predecessors:

$$M_{\text{local}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[j < i - k + 1] \vee \mathbf{1}[j > i].$$

Let s_i denote the speaker of utterance u_i . The inter-speaker mask blocks utterances produced by the same speaker, leaving only preceding utterances from other speakers accessible:

$$M_{\text{inter}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[s_i = s_j].$$

Conversely, the intra-speaker mask blocks utterances produced by other speakers and retains the target speaker’s preceding conversational history:

$$M_{\text{intra}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[s_i \neq s_j].$$

Thus, the four masks represent unrestricted historical context, local context, other-speaker context, and same-speaker context, respectively. For each $M_m \in \mathcal{M}$, the mask is applied to the shared attention scores S :

$$S_m(i, j) = \begin{cases} S(i, j), & M_m(i, j) = 0, \\ -\infty, & M_m(i, j) = 1. \end{cases}$$

The corresponding attention probabilities and contextual representations are

$$A_m = \text{softmax}(S_m), \quad Z_m = A_m V.$$

This computation is performed independently for all four masks using the same query, key, and value representations. The four outputs are then concatenated along the hidden dimension:

$$Z_{\text{concat}} = [Z_{\text{global}} \parallel Z_{\text{local}} \parallel Z_{\text{inter}} \parallel Z_{\text{intra}}] \in \mathbb{R}^{T \times 4H}.$$

Finally, a learned projection maps the concatenated representation back to the original hidden dimension:

$$Z = Z_{\text{concat}} W_P + b_P, \quad W_P \in \mathbb{R}^{4H \times H}.$$

The motivation was that these masks could encourage different attention branches to capture complementary information. For example, the local branch could emphasize short-range conversational continuity, while the inter-speaker and intra-speaker branches could capture information associated with interactions between participants and each speaker’s own history. The

global branch served as a less restrictive reference that retained access to the entire preceding conversation.

Despite the increased architectural structure and representational capacity, the resulting performance remained similar to that of the simpler attention model. I therefore examined whether the four branches were actually producing substantially different representations.

For two contextual representations X and Y , I measured their similarity using cosine similarity,

$$\text{CosSim}(X, Y) = \frac{\langle X, Y \rangle}{\|X\|_2 \|Y\|_2},$$

Jensen-Shannon divergence,

$$\text{JSD}(P \parallel Q) = \frac{1}{2}D_{\text{KL}}(P \parallel M) + \frac{1}{2}D_{\text{KL}}(Q \parallel M),$$

where

$$M = \frac{1}{2}(P + Q),$$

and the Frobenius norm,

$$\|X - Y\|_F = \sqrt{\sum_{i,j} (X_{ij} - Y_{ij})^2}.$$

The cosine similarity measures the directional similarity between representations, while the Jensen-Shannon divergence measures the difference between two probability distributions, and the Frobenius norm measures the overall magnitude of the difference between two matrices.

The representations produced by the different masks were generally highly similar. In particular, the global representation frequently exhibited high cosine similarity with representations produced by the other masks. Although some mask pairs showed greater divergence according to the three measures, such as between the listener and speaker masks, the overall results indicated substantial overlap between the information encoded by the different branches.

This suggested that the additional architectural structure appeared to provide more ways of representing the context without necessarily providing more task-relevant information. This offered a possible explanation for why the multi-mask architecture failed to produce a substantial improvement over the simpler attention module and the baseline.

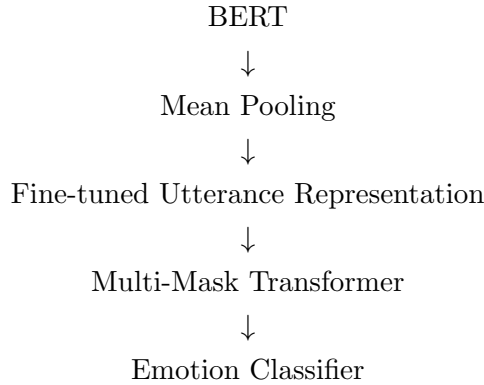
5 Final Methodology

5.1 Overview

Following the preceding experimental investigations, the final model combines a fine-tuned BERT utterance representation with a multi-mask Transformer encoder. The model is designed to preserve the semantic information learned at the utterance level while introducing conversation-level contextualization through several structurally different attention patterns.

The final training procedure is separated into two stages. First, BERT is fine-tuned using the mean-pooled representation of individual utterances for emotion classification. The resulting BERT representation is then used as the input to the conversation-level attention model, which is trained separately. This procedure allows the utterance representation and the conversation-level contextualization mechanism to be optimized independently.

The final architecture can therefore be summarized as



5.2 Fine-Tuned Utterance Representation

Let a conversation consist of T utterances:

$$\mathcal{C} = (u_1, u_2, \dots, u_T).$$

The model hidden dimension is denoted by $d = 768$. Each utterance u_t is independently tokenized and processed by BERT. If u_t contains n_t valid tokens, BERT produces the token-representation matrix

$$E_t = \begin{bmatrix} e_{t,1}^\top \\ e_{t,2}^\top \\ \vdots \\ e_{t,n_t}^\top \end{bmatrix} \in \mathbb{R}^{n_t \times d}, \quad e_{t,i} \in \mathbb{R}^d.$$

Instead of using only the [CLS] representation, the valid token representations are aggregated

through mean pooling:

$$x_t = \frac{1}{n_t} \sum_{i=1}^{n_t} e_{t,i}, \quad x_t \in \mathbb{R}^d.$$

The vector x_t represents the semantic content of utterance u_t .

During the first training stage, this representation is used directly for emotion classification. BERT is fine-tuned using the resulting classification objective, allowing its parameters to adapt to the emotion recognition task. After this stage, the resulting utterance encoder provides the representations used by the conversation-level model.

This stage serves as the foundation of the final architecture: the subsequent attention mechanism operates on representations that have already been adapted to the target emotion classification task rather than on an unmodified pretrained representation.

5.3 Conversation-Level Multi-Mask Attention

The utterance vectors are stacked row-wise to form the initial conversation-level representation:

$$X^{(0)} = \begin{bmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_T^\top \end{bmatrix} \in \mathbb{R}^{T \times d}.$$

The model contains $L = 8$ Transformer encoder layers. Let $X^{(\ell)} \in \mathbb{R}^{T \times d}$ denote the input to layer ℓ , where $\ell \in \{0, \dots, L-1\}$. Because the encoder uses pre-normalization, the attention input is

$$\bar{X}^{(\ell)} = \text{LayerNorm}\left(X^{(\ell)}\right).$$

Each layer uses $h = 8$ attention heads. The four masked contextual views share the same query, key, and value projections within a layer; the model therefore does not learn a separate attention module for each mask. The full projections at layer ℓ are

$$Q^{(\ell)} = \bar{X}^{(\ell)} W_Q^{(\ell)}, \quad K^{(\ell)} = \bar{X}^{(\ell)} W_K^{(\ell)}, \quad V^{(\ell)} = \bar{X}^{(\ell)} W_V^{(\ell)},$$

where

$$W_Q^{(\ell)}, W_K^{(\ell)}, W_V^{(\ell)} \in \mathbb{R}^{d \times d}.$$

The dimension of each attention head is

$$d_h = \frac{d}{h} = \frac{768}{8} = 96.$$

For head $a \in \{1, \dots, h\}$, let

$$Q_a^{(\ell)}, K_a^{(\ell)}, V_a^{(\ell)} \in \mathbb{R}^{T \times d_h}$$

denote the corresponding slices of $Q^{(\ell)}$, $K^{(\ell)}$, and $V^{(\ell)}$. The unmasked score matrix for head a is

$$S_a^{(\ell)} = \frac{Q_a^{(\ell)} \left(K_a^{(\ell)}\right)^\top}{\sqrt{d_h}} \in \mathbb{R}^{T \times T}.$$

These eight score matrices are computed once per encoder layer and reused across the four contextual masks.

5.4 Attention Mask Construction

The mask-index set is

$$\mathcal{M} = \{\text{global}, \text{local}, \text{inter}, \text{intra}\}.$$

For each $m \in \mathcal{M}$, the binary matrix $M_m \in \{0, 1\}^{T \times T}$ specifies whether query position i may attend to key position j . A value of 0 permits attention, whereas a value of 1 blocks it.

Before constructing the four contextual masks, a common base mask combines causal and padding constraints. The causal mask is

$$M_{\text{causal}}(i, j) = \mathbf{1}[j > i],$$

where $\mathbf{1}[\cdot]$ is the indicator function. Let $M_{\text{pad}}(i, j) = 1$ when key position j is padding and 0 otherwise. The shared base mask is

$$M_{\text{base}}(i, j) = M_{\text{causal}}(i, j) \vee M_{\text{pad}}(i, j).$$

Each of the four final masks is then constructed by adding a different structural restriction to this common base.

5.4.1 Global Mask

The global mask applies only the common causal and padding constraints:

$$M_{\text{global}}(i, j) = M_{\text{base}}(i, j).$$

Therefore, for a target utterance u_t , all valid preceding utterances, including the target utterance itself, remain accessible.

5.4.2 Local Mask

The local mask restricts attention to a fixed window of the most recent k utterances. For query position i and key position j ,

$$M_{\text{local}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[j < i - k + 1].$$

Because the causal constraint is already included in M_{base} , this effectively permits attention only to the target utterance and its preceding $k - 1$ utterances.

5.4.3 Inter-Speaker Mask

The inter-speaker mask restricts attention to utterances produced by the other speaker. Let s_i denote the speaker associated with utterance u_i . The additional restriction is

$$M_{\text{inter}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[s_i \neq s_j].$$

Thus, an utterance can attend only to preceding utterances produced by a different speaker.

5.4.4 Intra-Speaker Mask

Conversely, the intra-speaker mask restricts attention to utterances produced by the same speaker:

$$M_{\text{intra}}(i, j) = M_{\text{base}}(i, j) \vee \mathbf{1}[s_i = s_j].$$

Consequently, this attention pattern captures the target speaker’s own preceding conversational history.

5.4.5 Masked Attention and Feature Combination

At encoder layer ℓ , mask $m \in \mathcal{M}$ is applied independently to every head a . The masked score matrix is

$$\tilde{S}_{m,a}^{(\ell)}(i,j) = \begin{cases} S_a^{(\ell)}(i,j), & M_m(i,j) = 0, \\ -\infty, & M_m(i,j) = 1. \end{cases}$$

Applying a row-wise softmax gives the attention probabilities

$$A_{m,a}^{(\ell)} = \text{softmax}_j \left(\tilde{S}_{m,a}^{(\ell)} \right) \in \mathbb{R}^{T \times T}.$$

The output of head a under mask m is

$$O_{m,a}^{(\ell)} = A_{m,a}^{(\ell)} V_a^{(\ell)} \in \mathbb{R}^{T \times d_h}.$$

The eight head outputs for mask m are concatenated to produce one d -dimensional contextual view:

$$Z_m^{(\ell)} = \left[O_{m,1}^{(\ell)} \parallel \dots \parallel O_{m,h}^{(\ell)} \right] \in \mathbb{R}^{T \times d}.$$

The four contextual views are then concatenated:

$$Z_{\text{cat}}^{(\ell)} = \left[Z_{\text{global}}^{(\ell)} \parallel Z_{\text{local}}^{(\ell)} \parallel Z_{\text{inter}}^{(\ell)} \parallel Z_{\text{intra}}^{(\ell)} \right] \in \mathbb{R}^{T \times 4d}.$$

A learned projection maps the concatenated representation back to the model dimension:

$$U^{(\ell)} = Z_{\text{cat}}^{(\ell)} W_P^{(\ell)} + b_P^{(\ell)} \in \mathbb{R}^{T \times d},$$

where

$$W_P^{(\ell)} \in \mathbb{R}^{4d \times d}, \quad b_P^{(\ell)} \in \mathbb{R}^d.$$

The representation $U^{(\ell)}$ is the output of the multi-mask attention module at layer ℓ .

5.4.6 Transformer Encoder

Each multi-mask attention operation is embedded within a standard Transformer encoder layer. The layer uses pre-normalization followed by a residual connection:

$$Y^{(\ell)} = X^{(\ell)} + \text{Dropout}\left(U^{(\ell)}\right).$$

The feed-forward network uses an intermediate dimension $d_{\text{ff}} = 3072$. For a matrix $R \in \mathbb{R}^{T \times d}$,

$$\text{FFN}^{(\ell)}(R) = \text{GELU}\left(RW_1^{(\ell)} + b_1^{(\ell)}\right)W_2^{(\ell)} + b_2^{(\ell)},$$

where

$$W_1^{(\ell)} \in \mathbb{R}^{d \times d_{\text{ff}}}, \quad W_2^{(\ell)} \in \mathbb{R}^{d_{\text{ff}} \times d}.$$

The second pre-normalized residual update is

$$X^{(\ell+1)} = Y^{(\ell)} + \text{Dropout}\left(\text{FFN}^{(\ell)}\left(\text{LayerNorm}(Y^{(\ell)})\right)\right).$$

Repeating this update for $\ell = 0, \dots, L - 1$ produces the final conversation-level matrix $X^{(L)} \in \mathbb{R}^{T \times d}$.

5.4.7 Separated Training

The final system is trained in two stages.

In the first stage, the BERT encoder is fine-tuned using mean pooling and an emotion classification objective. This produces an utterance representation that has been adapted to the emotion recognition task.

In the second stage, the resulting utterance representations are provided to the multi-mask Transformer encoder. The conversation-level model is then trained to use contextual information from the conversation to improve emotion classification.

This separation distinguishes two learning problems: learning a useful representation of an individual utterance and learning how conversational context should modify that representation. The experimental investigation of this training strategy is discussed separately in the experimental results.

5.4.8 Final Prediction

The contextualized representation of utterance u_t is the t -th row of the final encoder output:

$$z_t = X_{t,:}^{(L)} \in \mathbb{R}^d.$$

For $c = 7$ emotion classes, the classifier produces logits and class probabilities

$$o_t = z_t W_c + b_c, \quad p_t = \text{softmax}(o_t),$$

where $W_c \in \mathbb{R}^{d \times c}$, $b_c \in \mathbb{R}^c$, and $p_t \in \mathbb{R}^c$. The predicted emotion label is

$$\hat{y}_t = \arg \max_{q \in \{1, \dots, c\}} p_{t,q}.$$

If $y_t \in \{1, \dots, c\}$ is the ground-truth emotion label for utterance u_t , the conversation-level cross-entropy loss is

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log p_{t,y_t}.$$

The final model therefore combines task-adapted utterance representations with several structurally constrained views of conversational context. The four attention patterns provide complementary contextual information, while their shared projections ensure that the differences between the resulting representations arise from the imposed contextual constraints rather than from independently parameterized attention modules.

6 Experimental Setup

6.1 Dataset and Task

The experiments are conducted on the DailyDialog dataset, a multi-turn dialogue dataset annotated with seven emotion categories, including Neutral. Each dialogue is treated as an ordered sequence of utterances, with a maximum of 36 turns considered during model processing.

The task is formulated as utterance-level emotion recognition in conversation (ERC). For each target utterance, the model predicts one of the seven emotion labels while having access only to the conversational context permitted by the corresponding model and attention mask.

The primary evaluation metric is macro-F1 over the six non-neutral emotion classes. Neutral is excluded from the primary metric because it is substantially more frequent than the other emotion categories and can otherwise dominate aggregate performance. Validation and test results are reported separately throughout the experiments.

6.2 Experimental Organization

The experiments are organized around explicit hypotheses rather than isolated architectural modifications. Each experiment records the hypothesis being tested, the configuration, the

architectural change, the training procedure, the resulting validation and test performance, observations, interpretation, and the next experimental step.

The experiments conducted during the research process are divided into two categories. Earlier experiments are exploratory investigations used to understand the behavior of different utterance representations, context aggregation mechanisms, attention structures, and auxiliary modifications. The final experiment is maintained as a fully reproducible experiment with its configuration, training procedure, and relevant evaluation information explicitly recorded.

The final experiment therefore serves as the reproducibility reference for the final methodology. The earlier exploratory experiments are retained through their available notes, logs, and configurations, but are not retroactively reconstructed as fully reproducible runs because doing so would require recovering multiple intermediate code states and configurations across the development history.

6.3 Utterance-Level Training

The utterance-level component uses `bert-base-uncased` as its pretrained language model. Each utterance is represented using mean pooling over the valid BERT token representations.

Before training the conversation-level model, the BERT-based utterance representation is fine-tuned separately for emotion recognition. The utterance-level model is trained for 5 epochs with a batch size of 16 and a learning rate of 1×10^{-5} . A dropout rate of 0.1 is used for the BERT component.

The resulting checkpoint is subsequently used to initialize the representation stage of the conversation-level model. This separate training stage provides the starting point for the final conversation-level experiment rather than using an entirely unadapted pretrained BERT representation.

6.4 Conversation-Level Training

The conversation-level model consists of the multi-mask Transformer architecture described in the Final Methodology. It uses a hidden size of 768, eight attention heads, a feed-forward dimension of 3072, eight Transformer encoder layers, and a dropout rate of 0.3.

The local attention mask uses a window size of 3. The model therefore evaluates the global, local, inter-speaker, and intra-speaker contextual views described previously.

The conversation-level model is trained for 30 epochs with a batch size of 4 and a learning rate of 1×10^{-4} . A weight decay of 0.01 is applied during optimization.

6.5 Separated Training and BERT Unfreezing

The final experiment uses a two-stage training procedure. First, the utterance-level BERT model is fine-tuned independently for five epochs using the utterance-level emotion recognition

objective. Its resulting parameters are then used to initialize the conversation-level training stage.

During conversation-level training, BERT is initially frozen for the first eight epochs. This prevents the newly initialized conversation-level components from immediately modifying the pretrained and task-adapted utterance representation while they begin to learn conversational contextualization.

After epoch 8, the designated upper portion of BERT is made trainable, allowing the utterance representation to adapt jointly with the conversation-level model during the remainder of training. In the final configuration, the conversation-level training stage uses a `bert_trainable_layer` setting of 4 after the initial freezing period.

Thus, the conversation-level training procedure can be summarized as

$$\underbrace{\text{Task-adapted BERT}}_{\text{separate utterance training}} \rightarrow \underbrace{\text{BERT frozen}}_{\text{epochs 1-8}} \rightarrow \underbrace{\text{partial BERT fine-tuning}}_{\text{remaining epochs}}.$$

This procedure is part of the final experimental configuration and is therefore distinguished from the earlier experiments that investigated alternative training strategies.

6.6 Optimization and Regularization

The final configuration uses separate learning rates for the major components of the model. The BERT component uses a learning rate of 1×10^{-5} , while the conversation-level attention component uses 1×10^{-4} . The utterance classification head uses 5×10^{-4} , and the conversation classification head uses 1×10^{-4} .

Dropout is set to 0.1 for the BERT component, 0.2 for the utterance-level classification head, and 0.3 for both the conversation-level attention component and conversation-level classification head.

The attention regularization mechanism investigated in an earlier experiment is disabled in the final configuration. Although the configuration retains an `attention_reg_rate` of 0.1, `use_attention_reg` is set to false, and therefore this term does not contribute to the final training objective.

6.7 Computational Configuration and Reproducibility

All experiments in the final reproducible run use random seed 42. CUDA acceleration is enabled, together with automatic mixed precision using BF16. The final configuration uses deterministic execution where supported by the underlying software and hardware environment.

The final experiment was executed using the configuration stored separately from the default configuration. This configuration specifies the dataset, model architecture, optimization parameters, training schedules, freezing and unfreezing behavior, precision settings, and reproducibility options.

The final experiment is intended to be directly reproducible from the recorded configuration and accompanying implementation. The complete implementation and the configuration required to reproduce the final experiment are provided with the project repository.

6.8 Final Experimental Configuration

The principal settings of the final reproducible experiment are summarized below.

Table 1: Final experimental configuration.

Component	Setting
Dataset	DailyDialog
Number of emotion classes	7
Maximum dialogue length	36 turns
Utterance encoder	BERT-base-uncased
Utterance representation	Mean pooling
Utterance training epochs	5
Utterance batch size	16
BERT learning rate	1×10^{-5}
Conversation training epochs	30
Conversation batch size	16
Attention hidden size	768
Attention heads	8
Transformer layers	8
Feed-forward dimension	3072
Local attention window	3
Attention dropout	0.3
Weight decay	0.01
Initial BERT freezing	8 epochs
BERT trainable layer setting	4
Attention regularization	Disabled
Random seed	42
Mixed precision	BF16
CUDA	Enabled
Deterministic run	Enabled
Primary metric	Non-neutral macro-F1

The same experimental configuration is used for the reported final results. Unless otherwise stated, changes appearing in the earlier experimental results refer to explicitly documented deviations from this final configuration.

7 Results

The final experiment was evaluated using two multi-mask attention configurations, alongside the single-utterance recognition baseline. The primary evaluation metric is macro-F1 over the six non-neutral emotion classes.

Table 2: Performance of the single-utterance baseline and final multi-mask attention configurations.

Configuration	Validation F1-m	Test F1-m
Single-utterance baseline	0.3506	0.4397
Multi-mask attention, $\text{lr} = 5 \times 10^{-5}$	0.3517	0.4400
Multi-mask attention, $\text{lr} = 1 \times 10^{-4}$	0.3595	0.4331

The second multi-mask configuration, which used a learning rate of 1×10^{-4} for both the attention module and the conversation classification head, achieved the highest validation F1-m of 0.3595. However, the first configuration, using a learning rate of 5×10^{-5} , achieved the highest test F1-m of 0.4400, marginally exceeding the single-utterance baseline (0.4397).

Although the second configuration improved validation performance, this improvement did not translate into higher test performance. The results therefore indicate a difference between the validation and test behavior of the two learning-rate configurations, without providing conclusive evidence of a consistent improvement over the single-utterance baseline.

8 Analysis & Discussion

The experiments did not demonstrate a consistent improvement over the single-utterance baseline. Although the multi-mask architecture achieved a comparable test performance, the results do not provide sufficient evidence that combining global, local, intra-speaker, and inter-speaker attention leads to a meaningful improvement in emotion recognition. The higher validation score obtained with the larger learning rate was not accompanied by better test performance, suggesting that the observed differences may be sensitive to optimization and generalization.

The results also accentuated the difficulty of evaluating a complex contextual architecture without controlled ablations. Since the final architecture and setup including the staged training pipeline and multi-mask self-attention is introduced at the same time, the effect of different components, fine-tuning procedure and other training choices were not isolated.

Despite the absence of a clear performance improvement, the project provided a reproducible comparison between single-utterance and conversation-level modeling. It also exposed practical challenges in designing contextual architectures, including optimization sensitivity, architectural complexity, and the need to separate architectural hypotheses from training configurations.

9 Lessons Learned

I entered the project with the assumption that Transformer-based architectures were the key to solving most Natural Language Processing problems, including emotion recognition. However, through experimentation, I found that architecture alone does not determine how a model behaves. Training procedures, hyperparameters, heuristics, and optimization choices can substantially affect the final performance of a model. This made me more cautious about attributing performance differences solely to architectural changes.

I also learned that research is a highly iterative process of investigating a problem through successive hypotheses and experiments. Many initial assumptions may not hold in practice, and experiments can reveal problems that were not anticipated at the beginning. As a result, the final methodology and conclusions do not necessarily have to align with the original expectations. In this project, several approaches that appeared reasonable initially performed poorly, while simpler approaches proved more competitive than expected.

Reproducibility is another important part of the research process. Although I conducted many experiments and retained their logs, configurations, and results, some earlier experiments were not stored together with the exact code version that produced them. Without the corresponding implementation, an experiment cannot be reliably reproduced, making it difficult to perform controlled comparisons or ablations later. This showed me that recording the code version, configuration, random seed, and relevant environment information is an indispensable part of research.

Finally, I learned that greater architectural complexity does not necessarily translate into better performance. In this project, the relatively simple BERT mean-pooling baseline achieved performance comparable to the substantially more complex Transformer-based architecture using four different attention masks. This reinforced the importance of establishing strong baselines and evaluating whether additional architectural complexity provides measurable benefits, rather than assuming that a more sophisticated design will inherently perform better.

10 Future Work

The research leaves several questions unresolved, providing potential directions for further investigation. These directions are presented as possible extensions of the current work rather than planned continuations of the project.

10.1 Analysis of Learned Attention Patterns

An analysis of the learned attention patterns could provide additional insight into how the model uses conversational context. Examining the attention distributions associated with the global, local, inter-speaker, and intra-speaker masks may help characterize the contextual information emphasized by each masking strategy. Such analysis could also investigate whether the attention patterns correspond to meaningful conversational relationships or emotional transitions.

10.2 Controlled Ablation of Attention Masks

Controlled ablation experiments could be conducted to isolate the contribution of individual attention masks. Removing or evaluating the global, local, inter-speaker, and intra-speaker masks independently would help determine whether they provide complementary information or introduce redundancy. These experiments could also clarify whether the combined architecture offers advantages over simpler contextual attention configurations.

10.3 Alternative Contextual Representations

Further work could investigate alternative methods of representing conversational context. Possible directions include different utterance-level pooling strategies, contextual aggregation mechanisms, and representations that explicitly model temporal or speaker-related relationships. Such comparisons could help distinguish the effects of the utterance representation from those of the conversation-level architecture.

10.4 Comparison with Other Conversation-Level Architectures

The multi-mask attention architecture could be compared with other conversation-level modeling approaches under a controlled experimental protocol. These comparisons may include recurrent architectures, standard Transformer-based contextualization, and other methods for modeling speaker interactions. A consistent evaluation framework would help clarify the relative behavior of the proposed masking strategy.

10.5 Analysis of Emotional Transitions

Another potential direction is to analyze errors associated with different emotional transitions. Rather than evaluating only aggregate emotion recognition performance, future investigations could examine whether the model behaves differently when emotions remain stable, change gradually, or transition between different categories. This analysis could provide a more detailed understanding of the relationship between conversational context and emotion recognition errors.

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